



in collaboration with



Title:

Lab 1 Task

Course:

ST5003CMD-Operating Systems, Security and Network

Institution:

Softwarica College of IT and E-Commerce

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1.File Operations:

```
sandip-rawal@sandip-rawal-VirtualBox: ~/linux_lab
sandip-rawal@sandip-rawal-VirtualBox:~/Desktop$ mkdir ~/linux_lab
sandip-rawal@sandip-rawal-VirtualBox:~/Desktop$ cd ~/linux_lab
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$
```

Figure 1 : Created and Navigated into a new tab directory.

```
sandip-rawal@sandip-rawal-VirtualBox: ~/Desktop
GNU nano 8.7.1 greeting.txt
Hello Linux!

sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ nano greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ cat greeting.txt
Hello Linux!
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ s
```

Figure 2 : Created and Edited a text file using nano editor

```
App Center sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ cp greeting.txt backup.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls
backup.txt  greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ss

Help sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ mv backup.txt old_greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls
greeting.txt  old_greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ s
```

Figure 3 : Copied the file and renamed the copy.

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ rm old_greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls
greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$
```

Figure 4 : Removed a File.

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ mkdir temp_dir
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls
greeting.txt  temp_dir
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ rmdir temp_dir
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls
greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$
```

Figure 5 : Created and removed Temporary Directory

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls -l greeting.txt
-rw-rw-r-- 1 sandip-rawal sandip-rawal 13 Sep  9 09:28 greeting.txt
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ S
```

Figure 6 : Permission Checked

2. APT Package Management

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [471 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [171 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.5 kB]
Get:8 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:9 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [601 kB]
Get:10 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:11 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [268 kB]
Get:12 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:13 http://np.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Fetched 2,299 kB in 4s (634 kB/s)
31 packages can be upgraded. Run 'apt list --upgradable' to see them.
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$
```

Figure 7 : Updated the local package

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ sudo apt install build-essential
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  build-essential

Installing dependencies:
  bzip2      g++-15-x86-64-linux-gnu  gcc-x86-64-linux-gnu  libcc1-0      libgcc-15-dev  libstdc++-15-dev
  dpkg-dev  g++-x86-64-linux-gnu    libalgorithm-diff-perl  libcrypt-dev  libhwasan0    libtsan2
  fakeroot  gcc                    libalgorithm-diff-xs-perl  libdpkg-perl  libitm1       libubsan1
  g++       gcc-15                 libalgorithm-merge-perl  libfakeroot   liblsan0      lto-disabled-list
  g++-15    gcc-15-x86-64-linux-gnu  libasan8                libfile-fcntllock-perl  libquadmath0  make
```

Figure 8 : Installed GCC and essential build tools

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Figure 9 : Verified (GCC installed)

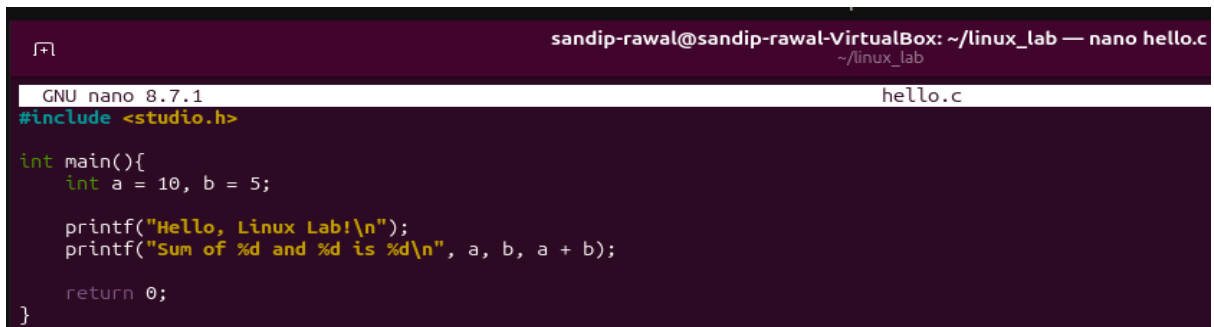
```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ apt search python 3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool
```

Figure 10 : Searched package repository(for another package)

3. C Compilation

Step 1: Write the Code



```

sandip-rawal@sandip-rawal-VirtualBox: ~/linux_lab — nano hello.c
~ /linux_lab
GNU nano 8.7.1 hello.c
#include <stdio.h>

int main(){
    int a = 10, b = 5;

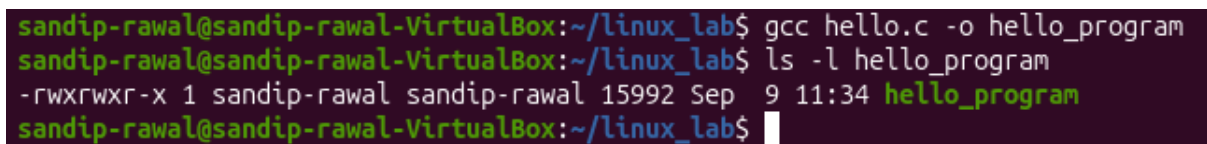
    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}

```

Figure 11 : C program(prints a greeting and sums two numbers)

Step 2 : Compile and Run

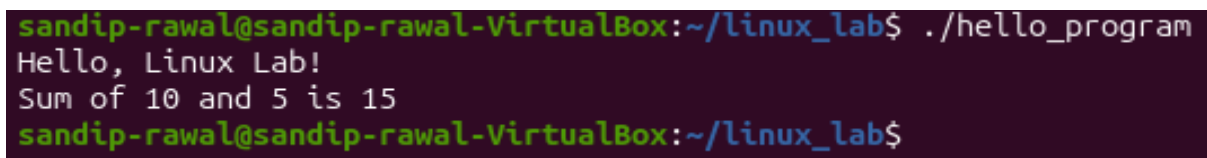


```

sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ gcc hello.c -o hello_program
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ls -l hello_program
-rwxrwxr-x 1 sandip-rawal sandip-rawal 15992 Sep  9 11:34 hello_program
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$

```

Figure 12 : Executed and Created(C source code and file)

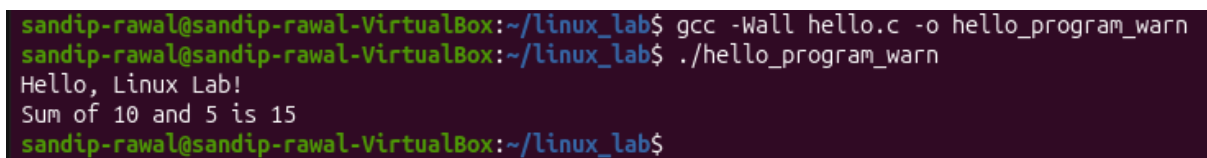


```

sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$

```

Figure 13 : Ran compiled program and got output



```

sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ gcc -Wall hello.c -o hello_program_warn
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$

```

Figure 14 : Warnings enabled

Step 3: Modify and Recompile

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ nano hello.c
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ gcc -Wall hello.c -o hello_program
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300
```

Figure 15 : Modified variables and Reran(got changes)

4. Cleanup and Verification

```
sandip-rawal@sandip-rawal-VirtualBox:~/linux_lab$ cd ~
sandip-rawal@sandip-rawal-VirtualBox:~$ ls -lh ~/linux_lab/
total 40K
-rw-rw-r-- 1 sandip-rawal sandip-rawal 13 Sep  9 09:28 greeting.txt
-rw-rw-r-- 1 sandip-rawal sandip-rawal 164 Sep  9 11:42 hello.c
-rwxrwxr-x 1 sandip-rawal sandip-rawal 16K Sep  9 11:43 hello_program
-rwxrwxr-x 1 sandip-rawal sandip-rawal 16K Sep  9 11:38 hello_program_warn
```

Figure 16 : Returned to home directory and Listed all lab files

```
sandip-rawal@sandip-rawal-VirtualBox:~$ cat ~/linux_lab/hello.c
#include <stdio.h>

int main(){
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

Figure 17 : Reviewed final version of C source file

```
sandip-rawal@sandip-rawal-VirtualBox:~$ rm -rf ~/linux_lab
sandip-rawal@sandip-rawal-VirtualBox:~$ ls ~/linux_lab
ls: cannot access '/home/sandip-rawal/linux_lab': No such file or directory
```

Figure 18 : Removed entire lab directory to cleanup

Summary:

In this lab, I practiced fundamental Linux command-line operations, package management with APT, and C program compilation.

File Operations: I created a working directory (`mkdir`, `cd`) and used `nano` to create and edit a text file. I verified its content with `cat`, then practiced copying and renaming files using `cp` and `mv`. I removed unwanted files and directories using `rm` and `rmdir`, and checked file permissions with `ls -l`.

APT Package Management: I updated the package index using `sudo apt update`, then installed the GCC compiler and build tools via `sudo apt install build-essential`. I confirmed the installation by checking the compiler version with `gcc --version`, and searched for an additional package (`python3`) using `apt search`.

C Compilation: I wrote a simple C program (`hello.c`) using `nano`, which printed a greeting and performed a basic sum of two integers. I compiled it using `gcc hello.c -o hello_program`, verified the executable was created with `ls -l`, and ran it using `./hello_program`. I also recompiled with the `-Wall` flag to enable compiler warnings, then modified the source code (changing variable values) and recompiled to confirm the changes reflected correctly in the output.

Cleanup and Verification: Finally, I navigated back to the home directory, listed the lab directory contents in detail (`ls -lh`), and reviewed the final version of `hello.c` using `cat` to confirm my work was saved correctly. I optionally removed the lab directory using `rm -rf` to complete the cleanup.